

Workbook: RDM Exercises

MORNING SESSION

Exercise 1: Start a Flemish DMP template in DMP online

Live demo – follow along

Exercise 2: Build a dataflow

Description: This exercise helps you visualize your research process by creating a schematic overview of all data types and steps involved. You'll identify physical and digital data and methods, and use this to complete the data table in your DMP.

End goal = schematic diagram of all research data and steps

1. *Data points (in/outputs): Blue*
2. *Steps (methods, analysis, processing,...): Yellow*

! Build with post-its so everything is easily changeable and movable.

Instructions:

In pairs

1. Use the following guiding questions to **quickly explain your project** to your partner.
It is enough to answer one per letter, **choose the one** that fits your research best.
 - a. What question where you trying to answer?
What do you want to figure out?
What do you want to improve?
What problem do you want to solve?
 - b. What is the research about?
What object are you studying?
What are you doing your research on?
 - c. How will you find out?
How will you achieve this?
What method(s) will you use?
2. Add what you have defined in question 1 **on post-its**
 - a. End point = (part of) publication (blue)
 - b. Start point = answer to question 1b (blue)
 - c. Add method = answer to question 1c (blue if data point, yellow if method/step). --> somewhere in between
3. **Select one or two paths** if you have a larger project. Make sure at least one path is described in detail.
4. Fill in gaps (steps/data points/decisions) in the right color.

*This is **not** a linear process: Write what you know and fill in as you go*

Not all materials necessary for a step/method would be considered as data (e.g. protocol, materials, ingredients,...) . Those don't need to be on your dataflow already. You can write those on a piece of scrap paper.

5. Discuss your data flow with your neighbor. Help each other to identify gaps.

If some steps seem unclear, use these guiding questions:

- a. "What is the input for this step?"
- b. "What is the output from this step"
- c. "What steps do I take with this datafile/sample?"
- d. "What steps will I take to get this datapoint?"

6. Update your data flow based on the discussion.

Exercise 3: Transfer the information from your data flow to the data table from your DMP template

Instructions:

1. Add the following information to all the data points in your dataflow (blue post-its):
 - a. Digital or Physical
 - b. New or Reused
 - c. If digital: format
 - d. Volume (number of samples OR files with expected size)
2. Translate this information to the data table from the DMP
3. Number the rows of your data table in the DMP, and add those to the post-its.

It is not always necessary to give each data point a separate row if there are a lot of similarities between the datapoints.

For example: all tables can be grouped together, all medical images, ...

Plenary Discussion

We will now reconvene with everyone. Please all share your thoughts and insights from this exercise in one or two sentences.

Questions for inspiration:

- How easy was it to make the dataflow?
- What insights did you gain from visualizing your dataflow?
- Did drawing the dataflow help you to complete the data table in your dmp?

AFTERNOON SESSION 1: Documentation at a micro scale – step by step

INTRO: paper airplane exercise

Live on slides

Exercise 4a: What elements are needed to document a step (yellow post-it) in your dataflow?

Guided group discussion:

Exercise 4b: What tools do you use to document each step (yellow post-it) in your dataflow?

Description: This exercise builds on the previous dataflow and helps you to build a documentation plan for your research data, with the goal of making it more reproducible.

Instruction

1. Discuss in pairs:
 - a. How do you currently document each step (what tool do you use)?
 - b. What documentation do you already create, and where are the gaps?
 - c. Do you use all elements from the previous exercise? If not, why? Would you change current practice?
2. Define what tool is the most appropriate for the “materials & methods” documentation of each step (ELN, ...). Write this on a pink post-it note and add it next to the step.
3. Add to these pink post-it notes which words you would use to describe the elements defined in exercise 4a.
4. Discuss this plan with your neighbor. Questions that can guide your discussion include:
 - a. Based on the provided documentation, would you be able to recreate a certain step of the data flow?
 - b. Did you include all elements from exercise 4a? If not, why?
 - c. Do you use similar words/terminology, or do the terms differ?
 - d. Do your documentation practices differ between wet lab and computational work?
 - e. Are there already templates available in your research group, and do they contain these elements? Do you see missing elements (from there or here)? Are there templates for both labwork and computerwork?

Plenary Discussion Questions:

- a. Have you encountered issues related to too little documentation and/or differences in terminology?
- b. Are there templates in your group? Is one person in charge of actively updating/streamlining these templates?
- c. How can we ensure consistency in documentation across team members?
- d. How can we make documentation less burdensome but still effective?
- e. What institutional or community support would help improve documentation?

Exercise 5: Beyond the steps – documentation for organization.

Description: Materials and methods are not the only documentation we need.

Guided group discussion

Instructions: Add the missing documentation for organization on pink post-its to your dataflow.

Take a global look at all the pink squares on your dataflow. Go to your **DMP** and summarize your documentation plan in section 2 (documentation and metadata). Add to section 2.2 if you have identified metadata as a tool to use.

Exercise 6: Where to store your data and the documentation?

Description: This exercise guides you in evaluating storage options for your data and documentation. You'll assess whether your current storage practices are secure, accessible, and appropriate for each data type.

Instructions:

1. Take your dataflow. Decide for all datapoints (blue squares):
 - a. Is the recommended storage place from my research group appropriate for this datapoint? (both physical and digital)
 - b. If not, where is the best place to store this?
 - c. Do I need additional security measures for this data?
 - d. Do I have enough space available (both physical and digital)?
2. Ask yourself for all documentation (Pink squares)
 - a. Does this documentation fit in the same storage location as my data?
 - b. If not, how will I solve this problem? (documentation is best kept in same location)
3. Take a look at section 3 from the DMP template (Data storage & back-up during the research project). Fill in what you can.

AFTERNOON SESSION 2: Documentation at a macro scale – whole project

INTRO: DUPLO exercise

Live on slides

Exercise 7: README file

Description: README.txt files can be used to document general information at the dataset or project level. They should be in an open format and it is recommended to use a template.

Instructions: Open README_ProjectTemplate.txt and fill in the GENERAL INFORMATION section with your project information.

Rename the file README_”yourproject”.txt

Exercise 8: File naming convention

Description: File naming conventions are a simple way to add order to your files and help to find them later. Rich and descriptive file names make it easier to search for files, understand at a glance what they contain, and tell related files apart. This exercise guides researchers through the process of creating a file naming convention for a group of related files.

Instructions: Fill in each section for a group of related files following the instructions – use the part of your project described in your dataflow. An example for microscopy files is provided. This exercise may be redone as needed, as different groups of files require different naming conventions.

Source: This exercise is based on the “File Naming Convention Worksheet” ([K. A. Briney, 2020a](#)).

Start with these questions (in pairs)

- What metadata do you usually include in file names?
- How do you handle versioning in your file names?
- What challenges have you faced with inconsistent file naming?

1. What group of files will this naming convention cover?

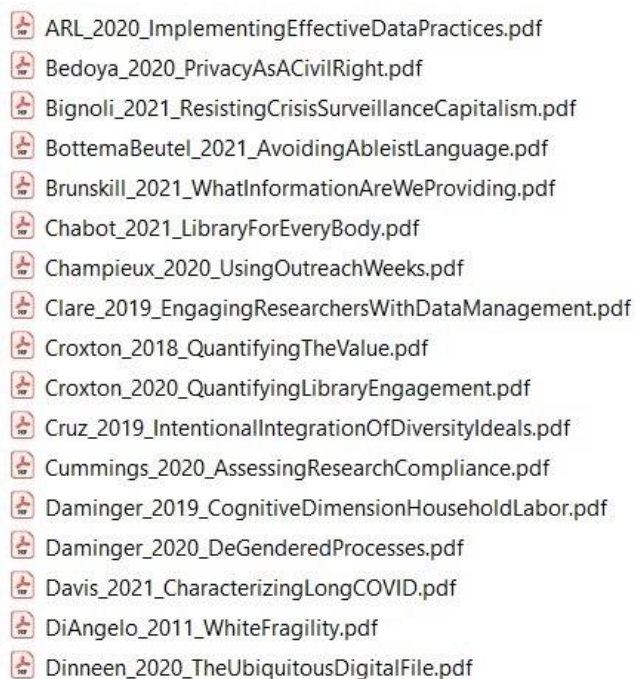
You can use different conventions for different file sets.

Think in this exercise about the part of your project you have built out in the dataflow.

Example: This convention will apply to all of my microscopy files, from raw image through processed image.

Commented [IM1]: From: Briney, K. (2023). *The Research Data Management Workbook*. Caltech Library. <https://doi.org/10.7907/z6czh-7zx60>

Met eventueel wat aanpassingen naar de onderzoekers hun eigen data, vind ik dit wel een heel fijne oefening.



ARL_2020_ImplementingEffectiveDataPractices.pdf
Bedoya_2020_PrivacyAsACivilRight.pdf
Bignoli_2021_ResistingCrisisSurveillanceCapitalism.pdf
BottemaBeutel_2021_AvoidingAbleistLanguage.pdf
Brunskill_2021_WhatInformationAreWeProviding.pdf
Chabot_2021_LibraryForEveryBody.pdf
Champieux_2020_UsingOutreachWeeks.pdf
Clare_2019_EngagingResearchersWithDataManagement.pdf
Croxtton_2018_QuantifyingTheValue.pdf
Croxtton_2020_QuantifyingLibraryEngagement.pdf
Cruz_2019_IntentionalIntegrationOfDiversityIdeals.pdf
Cummings_2020_AssessingResearchCompliance.pdf
Daminger_2019_CognitiveDimensionHouseholdLabor.pdf
Daminger_2020_DeGenderedProcesses.pdf
Davis_2021_CharacterizingLongCOVID.pdf
DiAngelo_2011_WhiteFragility.pdf
Dinneen_2020_TheUbiquitousDigitalFile.pdf

Example: Screenshot of article pdf's in a file system with consistent file names. File names use the convention FirstAuthorLastName_YEAR_ShortTitle.pdf

2. What information (metadata) is important about these files and makes each file distinct?

Ideally, pick three pieces of metadata; use no more than five. This metadata should be enough for you to visually scan the file names and easily understand what's in each one.

Example: For my images, I want to know date, sample ID, and image number for that sample on that date.

- 1.
- 2.
- 3.
- 4.
- 5.

3. Do you need to abbreviate any of the metadata or encode it?

If any of the metadata from step 2 is described by lots of text, decide what shortened information to keep. If any of the metadata from step 2 has regular categories, standardize the categories and/or replace them with 2- or 3-letter codes; be sure to document these codes.

Example: Sample ID will use a code made up of: a 2-letter project abbreviation (project 1 = P1, project 2 = P2); a 3-letter species abbreviation (mouse = “MUS,” fruit fly = “DRS”); and 3-digit sample ID (assigned in my notebook).

4. What is the order for the metadata in the file name?

Think about how you want to sort and search for your files to decide what metadata should appear at the beginning of the file name. If date is important, use ISO 8601-formatted dates (YYYYMMDD or YYYY-MM-DD) at the beginning of the file names so dates sort chronologically.

Example: My sample ID is most important so I will list it first, followed by date, then image number.

- 1.
- 2.
- 3.
- 4.
- 5.

5. What characters will you use to separate each piece of metadata in the file name?

Many computer systems cannot handle spaces in file names. To make file names both computer- and human-readable, use dashes (-), underscores (_), and/or capitalize the first letter of each word in the file names. A good convention is to use underscores to separate unrelated pieces of metadata and dashes to separate related pieces of metadata for parsing and readability.

Example: I will use underscores to separate metadata and dashes between parts of my sample ID.

6. Will you need to track different versions of each file?

You can track versions of a file by appending version information to end of the file name. Consider using a version number (e.g. “v01”) or the version date (use ISO 8601 format: YYYYMMDD or YYYY-MM-DD).

Example: As each image goes through my analysis workflow, I will append the version type to the end of the file name (e.g. “_raw,” “_processed,” and “_composite”).

7. Write down your naming convention pattern.

Make sure the convention only uses alphanumeric characters, dashes, and underscores. Ideally, file names will be 32 characters or less.

Example: My file naming convention is “SA-MPL-EID_YYYYMMDD_###_status.tif”

Examples are “P1-MUS-023_20200229_051_raw.tif” and “P2-DRS-285_20191031_062_composite.tif.”

8. Document this convention in your project README.txt and keep it with your files.

Pair discussion

- a. Would you be able to adapt your convention to the rest of your project?
- b. Could you use a similar method to decide on a naming scheme for your ELN?
What are the similarities and differences?

Exercise 9: Folder structure

Description: This exercise helps you design a logical folder structure for your project. A clear structure improves data findability and supports collaboration and long-term preservation. Implementing a file organization system is the first step toward creating order for your research data. Well-organized files make it easier to find the data you need without spending lots of time searching your computer. Every researcher organizes their files slightly differently, but the actual organizational system is less important than having a place where all of your files should logically go. This exercise prompts you to brainstorm organizational groupings and hierarchies to come up with an order for managing your research data.

Helpful resource: <https://www.kuleuven.be/rdm/en/guidance/data-standards/how-organise-data>

Instructions:

1. Go back to your dataflow. Find all the datapoints (blue post-its) that are digital, but leave them on the dataflow.
2. Copy them onto other blue post-it notes. Add whether it is considered part of “raw data”, “processed data” or “analysis”
3. Add new post-it notes with the following on them:
 - a. Literature
 - b. Articles
 - c. Grant documents
 - d. Figures
 - e. Conferences
4. Check your yellow and pink post-its for other things that are files (code, separate documentation files, ...). Also write these on new post-it notes in the correct color.
5. Use these post-its with their classification to create a folder structure. Add folder levels as needed. Adjust placement and hierarchies until you are happy with the organizational system you developed.
6. Document this folder structure in the **README.txt** and keep it with your files in the top level folder of your project.

Commented [IM2]: KU Leuven template:
https://www.kuleuven.be/rdm/en/guidance/documentation-metadata/README_template

Pair Discussion Questions:

- a. How do you currently organize your project folders?
- b. Would the other parts of your project fit into this folder structure? How would you need to adapt this?
- c. What folder structures have worked well (or not) in past projects?
- d. How do you handle shared folders in collaborative projects?

Exercise 10: Link with DMP

Add the information about your README file, including file-naming and folder structure, to section 2 in your DMP.

END OF EXERCISES:

Plenary Discussion Questions:

- a. Should naming conventions be flexible or strictly enforced?
- b. How can we make folder structures intuitive for new team members?
- c. How can we deal with changes in the project that affect the naming convention or folder structure?
- d. How can we make a uniform naming scheme across both the files and the ELN?
- e. How can we encourage consistent naming across a team?